University of Minnesota custom Azure course

Session 3 Participant Workbook

Data Protection in Both Directions

Azure Storage, Data Protection, and Resilience at Scale

Session: Day 2, Session 3, four hours

Use: Checkpoint discussion, design exercise or lab debrief, discussion prompts, knowledge check

Keep: Your worksheets are the input to the Session 6 roadmap capstone

All scenarios use synthetic data modeled on a large public research university. No real University of Minnesota data is used.

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1. Session map

Day 2, Session 3: Data Protection in Both Directions. Four hours including a fifteen-minute break.

Block	Minutes	Activity
Block 1	50	Azure Backup architecture and vault selection
Block 2	55	Ransomware resilience controls and the live attack demonstration
Break	15	
Block 3	45	On-premises to Azure and cloud workload protection
Block 4	75	Hands-on lab and debrief

1.1 What this session covers

Leadership wants the university to be able to say, with confidence, that it offers immutable data protection. This session covers both directions: Azure as a target for on-premises backups (MARS agent, Microsoft Azure Backup Server, offline seeding, passphrase custody, third-party platforms writing to immutable Blob) and protection of the workloads that will live in Azure (operational and vaulted blob backup, vaulted backup for Data Lake Storage, Azure Files, virtual machines, managed disks).

Between the two directions sit the controls that make immutability real against a compromised administrator: immutable vaults (reversible and irreversible, with WORM-backed storage where available), soft delete and enhanced soft delete, multi-user authorization with Resource Guard, built-in security alerts, and least-privilege backup roles. The vault choice (Recovery Services vault or Backup vault) is made by datasource, and the single management plane is Resiliency in Azure, which extends Azure Business Continuity Center.

1.2 Have ready

- Your current backup topology in mind: what backs up the file servers, the application servers, and the databases today; where the copies go; who can delete them.
- The Session 2 workbook: the archival landing zone is the pattern the third-party backup target reuses.
- A terminal or Cloud Shell signed in to your lab subscription for Block 4, with the `dataprotection` extension installed.

1.3 Materials you will receive

- Session 3 deck; demo script `S03_Demo_Script_Ransomware_Attack_Defeated`
- `S03_Lab_Guide_Immutable_Backup_Vault_MUA` (participants) and `S03_Lab_Solution_Immutable_Backup_Vault_MUA` (instructor)

- Protection pattern reference card (which vault, which control, which workload) and the executive briefing one-pager
- This workbook (participants) and the facilitator guide (instructor)

2. Block 1 checkpoint: choosing the vault

At the end of Block 1 the facilitator asks three questions about vault selection and boundaries. Answer for the university's own workloads.

Question 1

List the university workloads that go to a Recovery Services vault and those that go to a Backup vault. What decides it?

Question 2

A vault protects datasources in its own region and, for most workloads, its own subscription. How many vaults does the university need, and how would you name and tag them?

Question 3

Backup storage redundancy is set before the first item is protected. Which vaults get GRS with cross-region restore, and which stay LRS or ZRS?

3. Debrief design sprint: protection in both directions for your estate

The lab built a Backup vault that refused its own administrator. In the twenty-minute debrief, each group maps the university's real protection flows: three on-premises sources that will back up to Azure and three Azure workloads that will be protected in Azure, with the vault, the controls, and the separation of duties for each.

Use the lab's results as evidence: which operation was refused, what the message said, how long the first backup took.

3.1 Steps

1. Record your lab observations in Worksheet 1.
2. Complete Worksheet 2 for three on-premises sources (direction 1: on premises to Azure).
3. Complete Worksheet 3 for three Azure workloads (direction 2: protecting what lives in Azure).
4. Complete Worksheet 4: who owns what, and where the Resource Guard lives.
5. Be ready to state what you would tell your vice president 'immutable' means.

3.2 Worksheet: lab evidence

Observation	Your result
Protected operation attempted and the first line of the refusal	
Time from on-demand backup start to completion	
Restore target and result	

Observation	Your result
Something that surprised you	

3.3 Worksheet: direction 1, on premises to Azure

Decision	Source 1	Source 2	Source 3
Source (file server, application, database, backup platform image)			
Mechanism: MARS, MABS, third-party platform to immutable Blob			
Target: vault (type, redundancy) or Blob container (tier, WORM interval, lock)			
Seeding and bandwidth (Data Box, throttling)			
Passphrase or key custody			

3.4 Worksheet: direction 2, Azure workloads

Decision	Workload 1	Workload 2	Workload 3
Workload (VM, Blob flat, Data Lake, Azure Files, disk, database)			
Vault type and backup tier (operational, vaulted, both)			
Policy: frequency, retention, redundancy			
Controls: immutability (unlocked or locked), soft delete setting, MUA			
Restore test cadence and alternate location			

3.5 Worksheet: separation of duties

Role	Who holds it (team)	What they can do	What they cannot do
Backup Operator (daily work)			

Role	Who holds it (team)	What they can do	What they cannot do
Backup Contributor			
Resource Guard owner (Reader on the guard)			
Lock decision for Blob WORM policies			
Alert recipients			

3.6 What you present

- Three sources and three workloads with vault, tier, controls, and restore cadence.
- A separation-of-duties table with real team names (no individuals).
- One sentence for the vice president.

3.7 How your design will be critiqued

Lens	Meets	Exceeds
Reliability	Every workload has a vault, a policy, and a restore cadence	Restore to an alternate location named; evidence kept for audit
Security	Immutability, soft delete, and MUA assigned to every vault; passphrase custody named	Resource Guard in a separate subscription or tenant; least-privilege roles per team
Cost optimization	Retention length and vault redundancy justified	Operational versus vaulted tier chosen per data class with cost stated
Operational excellence	Azure Policy named as the enforcement of backup, immutability, and soft delete	Alerts routed to action groups owned by Security

Lens	Meets	Exceeds
Performance efficiency	Seeding and throttling considered for large initial backups	Data Box seeding sized to the initial backup

4. Discussion prompts

Work through these in your group or with a neighbor during the time the facilitator calls out. Write enough that you can defend the position in the room.

Prompt 1

Assume the administrator is compromised. Walk through the five attack paths from the demo against your current on-premises backup system. Which ones succeed today?

Block 2.

Prompt 2

MARS and MABS encrypt with a passphrase Microsoft cannot recover. Where does the university keep it, who can read it, and how is that tested?

Block 3.

Prompt 3

Cool and Archive as backup targets for third-party platforms: when does the cheaper tier cost more?

Block 3.

5. Knowledge check

Six questions, one best answer each. Mark your answer; the facilitator reviews them at the end of the session.

1. Which datasource is protected by a Backup vault rather than a Recovery Services vault?

- A. Azure VM
- B. Azure Files
- C. Azure Blob (vaulted backup)
- D. SQL Server in an Azure VM

2. What does enhanced soft delete add over standard soft delete?

- A. Immutability
- B. A configurable retention period up to 180 days and an always-on setting that cannot be turned off
- C. Cross-region restore
- D. Multi-user authorization

3. Where should the Resource Guard be placed?

- A. In the same resource group as the vault
- B. In a subscription or tenant owned by a different team than the vault administrators
- C. Inside the Backup vault
- D. In the Microsoft-managed subscription

4. What is the maximum retention of blob operational backup?

- A. 30 days
- B. 90 days
- C. 360 days
- D. 10 years

5. Which statement about hierarchical namespace accounts is correct?

- A. They support operational blob backup
- B. They are protected with vaulted backup for Data Lake Storage only
- C. They cannot be backed up
- D. They require a Recovery Services vault

6. An unlocked container-level WORM policy does which of the following?

- A. Blocks blob deletes but not storage account deletion
- B. Blocks storage account deletion
- C. Blocks reads
- D. Nothing until it is locked

6. Takeaways and the capstone

- Choose the vault by datasource; set redundancy before protecting items; plan vault boundaries by region, ownership, and workload class.
- Immutability, enhanced soft delete, and multi-user authorization together resist a compromised administrator; alerts and least-privilege roles make the attempts visible and rare.
- Protect both directions, keep the copy outside the blast radius, and put the Resource Guard, not the vault, in the separate subscription or tenant.
- A backup is only as good as its last tested restore; restore to an alternate location and keep the evidence.

> **Capstone input from this session**
>
> Immutable backup target for on-premises protection.
>
> Next: Session 4, resilience, disaster recovery, and multi-region design: what redundancy guarantees and what it does not, Site Recovery, and the RPO and RTO design method.

7. Notes

Your notes